

ASCE 7 Wind Load Calculations (Directional Procedure)

July 26, 2026

1 Design Parameters

The following fundamental parameters and building dimensions were utilized for the ASCE 7 wind load calculations:

Parameter	Value
Building Classification	Risk Category IV
Basic Wind Speed (m/s)	61.11
Enclosure Classification	Partially Enclosed Buildings
Exposure Category	B
Wind Directionality Factor (K_d)	0.85
Topographic Factor (K_{zt})	1.00
Ground Elevation Factor (K_e)	1.00
Gust Effect Factor	0.85
Tentative Velocity Pressure (K_z Pa)	1945.90
GCpi	+0.55 / -0.55
L (m)	40.00
B (m)	40.00
Direction of Ridge	B
Heights to Consider (m)	8.0, 10.0, 15.0, 18.0, 19.0
Eave Height (m)	20.00
Apex Height (m)	25.00

2 Velocity Pressure Profile (q_z)

The velocity pressure, evaluated at height z , is calculated using the following ASCE 7 formula:

$$q_z = 0.613 K_z K_{zt} K_d K_e V^2$$

- q_z : Velocity pressure (N/m^2 or Pa)
- V : Basic wind speed (m/s)
- K_d : Wind directionality factor
- K_e : Ground elevation factor
- K_{zt} : Topographic factor
- K_z : Velocity pressure exposure coefficient

2.1 Velocity Pressure Profile Table

Height (m)	Type	K_z	q_z (Pa)
8.0	User Input	0.671	1305.697
10.0	User Input	0.717	1395.208
15.0	User Input	0.807	1570.339
18.0	User Input	0.850	1654.012
19.0	User Input	0.862	1677.363
20.0	Eave Height	0.874	1700.714
22.5	Mean Roof Height	0.905	1761.036
25.0	Apex Height	0.936	1821.359

3 Pressure Coefficients (Cp)

3.1 Wall Pressure Coefficient (Cp) Table

Surface	Wind direction	L/B	Cp
Windward wall	All	All	0.8
Leeward wall	Normal to ridge	1.0	-0.5
Leeward wall	Parallel to ridge	1.0	-0.5
Side wall	All	All	-0.7

3.2 Roof Pressure Coefficient (Cp) Table - Normal Wind Direction

Surface	10°	14.0°	15°
Windward roof	-0.95	-0.78	-0.74
	-0.18	-0.18	-0.18
Leeward roof	-0.53	-0.51	-0.51

3.3 Roof Pressure Coefficient (Cp) Table - Parallel Wind Direction

Surface	Horizontal Distance from Windward Edge	h/L = 0.56
Roof (Parallel)	0 to h/2	-0.95, -0.18*
	h/2 to h	-0.88, -0.18*
	h to 2h	-0.53, -0.18*
	>2h	-0.35, -0.18*

4 MWFRS Pressure Summary

Design wind pressures for the Main Wind-Force Resisting System (MWFRS) are determined in accordance with ASCE 7 using the following equation:

$$p = qGC_p - q_i(GC_{pi})$$

- p : Design wind pressure (Pa)
- q : Velocity pressure (Pa), evaluated at height z for windward walls, or at height h for leeward/side walls and roofs
- q_i : Internal velocity pressure (Pa), evaluated at mean roof height h
- G : Gust-effect factor
- C_p : External pressure coefficient
- GC_{pi} : Internal pressure coefficient

4.1 MWFRS Pressure Summary Table - Normal Wind Direction

Surface	z (m)	q (Pa)	G	C_p	Net (+ GC_{pi})	Net (- GC_{pi})
Windward wall	8	1305.7	0.85	0.8	-80.7	1856.44
	10	1395.21	0.85	0.8	-19.83	1917.31
	15	1570.34	0.85	0.8	99.26	2036.4
	18	1654.01	0.85	0.8	156.16	2093.3
	19	1677.36	0.85	0.8	172.04	2109.18
	20	1700.71	0.85	0.8	187.92	2125.06
Leeward wall	All	1761.04	0.85	-0.5	-1717.01	220.13
Side walls	All	1761.04	0.85	-0.7	-2016.39	-79.25
Windward roof*	—	1761.04	0.85	-0.78	-2136.14	-199.0
	—	1761.04	0.85	-0.18	-1238.01	699.13
Leeward roof	—	1761.04	0.85	-0.51	-1731.98	205.16

4.2 MWFRS Pressure Summary Table - Parallel Wind Direction

Surface	z (m)	q (Pa)	G	C_p	Net (+ GC_{pi})	Net (- GC_{pi})
Windward wall	8	1305.7	0.85	0.8	-80.7	1856.44
	10	1395.21	0.85	0.8	-19.83	1917.31
	15	1570.34	0.85	0.8	99.26	2036.4
	18	1654.01	0.85	0.8	156.16	2093.3
	19	1677.36	0.85	0.8	172.04	2109.18
	20	1700.71	0.85	0.8	187.92	2125.06
	22.5	1761.04	0.85	0.8	228.93	2166.07
	25	1821.36	0.85	0.8	269.95	2207.09
Leeward wall	All	1761.04	0.85	-0.5	-1717.01	220.13
Side walls	All	1761.04	0.85	-0.7	-2016.39	-79.25
Roof (Roof (0 to h/2))	—	1761.04	0.85	-0.95	-2390.61	-453.47
	—	1761.04	0.85	-0.18	-1238.01	699.13
Roof (h/2 to h)	—	1761.04	0.85	-0.88	-2285.82	-348.69
	—	1761.04	0.85	-0.18	-1238.01	699.13
Roof (h to 2h)	—	1761.04	0.85	-0.53	-1761.92	175.22
	—	1761.04	0.85	-0.18	-1238.01	699.13
Roof (>2h)	—	1761.04	0.85	-0.35	-1492.48	444.66
	—	1761.04	0.85	-0.18	-1238.01	699.13